

MathQuestion-001

1 Question:

$$\lim_{n \rightarrow \infty} \frac{\log_n(n!)}{\sqrt{n?}}$$

1.1 Method1

We need to know:

Stirling's approximation

$$n! = n(n-1)\dots 2 \cdot 1 \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n, n \rightarrow \infty$$

Triangular number:

$$n? = n + (n-1) + \dots + 2 + 1 = \frac{n(n+1)}{2}$$

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{\log_n(n!)}{\sqrt{n?}} &= \lim_{n \rightarrow \infty} \frac{\sqrt{2} \ln(\sqrt{2\pi n} \cdot \left(\frac{n}{e}\right)^n)}{\ln n \sqrt{n^2 + n}} \\ &= \sqrt{2} \cdot \lim_{n \rightarrow \infty} \frac{\frac{1}{2} \ln(2\pi n) + n \ln n - n \ln e}{\ln n \cdot n \sqrt{1 + \frac{1}{n}}} \\ &= \sqrt{2} \left[\lim_{n \rightarrow \infty} \frac{\frac{1}{2} \ln(2\pi) \xrightarrow{0}}{\ln(n) n \sqrt{1 + \frac{1}{n}}} + \lim_{n \rightarrow \infty} \frac{n \ln n \xrightarrow{1}}{n \ln n \sqrt{1 + \frac{1}{n}}} - \lim_{n \rightarrow \infty} \frac{1 \xrightarrow{0}}{\ln n \sqrt{1 + \frac{1}{n}}} \right] \\ &= \sqrt{2} [0 + 1 - 0] \\ &= \sqrt{2} \end{aligned}$$

1.2 Method2

$$\begin{aligned}
\lim_{n \rightarrow \infty} \frac{\log_n(n!)}{\sqrt{n?}} &= \lim_{n \rightarrow \infty} \frac{\ln(n!)}{\sqrt{\frac{n(n+1)}{2}} \ln(n)} \\
&= \lim_{n \rightarrow \infty} \frac{\sqrt{2}}{\sqrt{n(n+1)} \cdot \ln(n)} \left(\ln \frac{n}{\textcolor{red}{n}} + \ln \frac{n-1}{\textcolor{red}{n}} + \dots + \ln \frac{1}{\textcolor{red}{n}} + \textcolor{red}{n \ln n} \right) \\
&= \lim_{n \rightarrow \infty} \frac{\sqrt{2}}{\sqrt{\frac{n(n+1)}{\textcolor{red}{n}^2}} \cdot \ln(n)} \left(\sum_{k=1}^n \ln \frac{k}{n} \cdot \frac{1}{\textcolor{red}{n}} \rightarrow \int_0^1 \ln x dx = -1 \right) + \lim_{n \rightarrow \infty} \frac{\sqrt{2} \cdot \cancel{n \ln n}^{\infty \rightarrow 1}}{\cancel{\sqrt{n(n+1)} \cdot \ln(n)}^{\infty \rightarrow 1}} \xrightarrow{\quad} \sqrt{2} \\
&= 0 + \sqrt{2} \\
&= \sqrt{2}
\end{aligned}$$